Rep. 4 equiprobable exits: $\frac{1}{4}$.

5. Red thread · Method [ALG] Program: choose a number; raise squared; add 1. Result for -3? Can we get 0? Justify.

Rep. For -3 : $(-3)^2+1=10$. No: $x^2 \geqslant 0$ therefore $x^2+1 \geqslant 1 > 0$.

Form 29 Mon. 11 Jan. 2027 Data grouped into classes: histogram and cumulative frequencies

1. Warming · Flash [STA] Give amplitude and center of class $[20;35[$.

Rep. Amplitude 15; centre $\frac{20+35}{2} = 27{,}5$.

2. Heart · Table [LEC] Classes $[0;10[$, $[10;20[$, $[20;30[$ The number of personnel is 4, 6, 10. increasing cumulative frequencies.

Rep. Total 20; frequency 20%, 30%, 50%; cumulative 20%, 50%, 100%.

3. Heart + · Flash [LEC] In a histogram with classes of the same amplitude, two rectangles have for heights 3 cm and 6 cm. What is the ratio of their staff?

Rep. The heights are proportional to the number of staff: 1 for 2.

4. Back · Flash [NUM] Write 0.00032 in scientific notation.

Rep. $3{,}2 \times 10^{-4}$.

5. Red thread · Flash [EQ] Solve in $\mathbb{R}$: $2(x+3) = 5x-9$.

Rep. $x = 5$

Trap: forget to distribute the 2 on the 3.

Form 30 March 12 Jan 2027 Weighted average, median class; spreadsheet and Python

1. Warming · Flash [CAL] Calculate $\frac{2\times 8+3\times 12}{5}$.

Rep. $\frac{52}{5} = 10{,}4$.

2. Heart · Method [STA] Classes $[0;10[$, $[10;20[$, $[20;30[$ 4 ; 7 ; 9. Estimate the average with class centres; give the middle class.

Rep. Estimated average $\frac{4\times 5+7\times 15+9\times 25}{20} = 17.5$; cumulative 4; 11; 20: 10 ^{e} and 11^{e} values are in $[10;20[$: middle class.

3. Heart + · Code reading [ALG] What does this program calculate and what value does it obtain?

```
Notes = [12, 15, 9, 14]
s = 0
for n in notes:
    s = s + n
print(s / len(notes))
```

Rep. Average scores: $\frac{50}{4} = 12{,}5$.

4. Back · Flash [PRO] A 45,000 Ar item is sold to $-$What is the sale price?

Rep. $45\,000 \times 0{,}70 = 31{,}500$ Ar.

5. Red thread · Prove [LIT] Program: choose n ; add 1 ; raise squared ; subtract n^2. Show that the result is always an odd number.

Rep. $(n+1)^2 - n^2 = 2n+1$: odd for the integer n.

Algebraic calculation, equations and inequities

Form 31 14 Jan. 2027 Relative integers

1. Warming · Flash [Sighs] Calculate 2^{-3}, 10^{-2} and $(-3)^2$.

Rep. $\frac{1}{8}$; 0,01 ; 9.

2. Heart · Method [Sighs] Simplify $\frac{2^3 \times 2^{-5}}{2^{-1}}$.

Rep. $\frac{2^{-2}}{2^{-1}} = 2^{-2+1} = 2^{-1} = \frac{1}{2}$.

3. Heart + · Method [Sighs] For $x \neq 0$, write $(x^2)^{-3} \times x^7$ in the form x^n.

Rep. $x^{-6} \times x^7 = x^1 = x$.

4. Back · Flash [STA] Median Series 4, 7, 9, 10, 12, 15?

Rep. Equal number: $\frac{9+10}{2} = 9{,}5$.

5. Red thread · Recognize [SOL] What solid has a disc-shaped base, a curved lateral surface and a One summit?

Rep. A cone of revolution.

Form 32 Mon. 18 Jan 2027 Square roots; $\sqrt{a^2} = |a|$

1. Warming · Flash [CAL] Calculate $\sqrt{36}$ and $\sqrt{0{,}09}$.

Rep. 6 ; 0,3.

2. Heart · Method [CAL] Write $\sqrt{72}$ in form $a\sqrt{2}$ and calculate $\sqrt{(-5)^2}$.

Rep. $\sqrt{72} = \sqrt{36 \times 2} = 6\sqrt{2}$; $\sqrt{(-5)^2} = |-5| = 5$.

3. Heart + · True/false [CAL] True or false? $\sqrt{9+16} = \sqrt{9} + \sqrt{16}$.

Rep. Wrong $\sqrt{25} = 5$ but $3 + 4 = 7$. Only the product is "separate": $\sqrt{ab} = \sqrt{a}\sqrt{b}$.

4. Back · Flash [Sighs] Calculate $(3^2)^3$.

Rep. $3^{2\times 3} = 3^6 = 729$.

5. Red thread · True/false [DEM] True or false? Justify. The diagonals of a diamond are perpendicular.

Rep. True. Ownership of the diamond (its diagonals are also mediating one another).

Form 33 Mar. 19 Jan. 2027 Fractional expressions; developed, reduced or factorized forms

1. Warming · Flash [CAL] Calculate $\frac{1}{2} + \frac{1}{3}$.

Rep. $\frac{5}{6}$.

2. Heart · Method [LIT] Write as a single fraction $\frac{1}{x} + \frac{1}{x+1}$ (for $x \neq 0$ and $x \neq -1$).

Rep. $\frac{x+1+x}{x(x+1)} = \frac{2x+1}{x(x+1)}$.

3. Heart + · Method [DEV] Factoring $x^2 - 4$ then simplify $\frac{x^2-4}{x-2}$ for $x \neq 2$.

Rep. $x^2 - 4 = (x-2)(x+2)$; therefore $\frac{x^2-4}{x-2} = x+2$.

The choice of form depends on the objective: factored to simplify, developed to compare.

4. Back · Flash [SUB] Calculate $2a - 3b$ for $a = 2$ and $b = -1$.

Rep. $4 + 3 = 7$.

5. Red thread · Flash [REP] ABCDEFGH is a straight pavement with $AB = 4$, $AD = 3$, $AE = 2$. In the coordinate system $(A; B, D, E)$, what are the coordinates of G ?

Rep. $G(4; 3; 2)$: G is the summit opposed to A.

A third coordinated approach appears in space.

Form 34 game. 21 Jan 2027 Inequality: sum, multiplication by a real

1. Warming · Flash [CMP] Complete by $<$ or $>$: $-3 \ldots -2$; $\frac{1}{3} \ldots 0{,}3$.

Rep. $-3 < -2$; $\frac{1}{3} > 0{,}3$ ($0{,}333\ldots > 0{,}3$).

2. Heart · Method [EQ] We have $a < b$. Compare $-2a$ and $-2b$ then $a+5$ and $b+5$.

Rep. $-2a > -2b$ (multiply by a negative reverses inequality); $a+5 < b+5$.

3. Heart + · Method [EQ] Knowing $2 < x < 3$ and $1 < y < 4$, frame $x+y$ then $x-y$.

Rep. $3 < x+y < 7$; for $x-y$: $-y \in]-4;-1[$ so $-2 < x-y < 2$.

Trap: removing member-to-member inequalities.

4. Back · Flash [CAL] Calculate $(-3) \times (-4) + 2 \times (-5)$.

Rep. $12 - 10 = 2$.

5. Red thread · Problem [IVD] A 120 cm by 84 cm plate is cut into identical squares, without loss. What's the biggest possible side?

Rep. PGCD $(120\,;\,84) = 12$: 12 cm squares on the side ($10 \times 7 = 70$ squares).

Form 35 Mon. 25 Jan 2027 Comparison by difference and quotient

1. Warming · Flash [CMP] Compare $\frac{2}{3}$ and $\frac{3}{4}$ by studying the sign of difference.

Rep. $\frac{2}{3} - \frac{3}{4} = -\frac{1}{12} < 0$: $\frac{2}{3} < \frac{3}{4}$.

2. Heart · Method [LIT] Compare x^2+1 and $2x$ for all real x.

Rep. $x^2 + 1 - 2x = (x-1)^2 \geqslant 0$: $x^2 + 1 \geqslant 2x$ equality if and only if $x = 1$.

3. Heart + · Method [CMP] Compare $\frac{7}{5}$ and $\frac{8}{6}$ by quotient (positive numbers).

Rep. $\frac{7/5}{8/6} = \frac{42}{40} > 1$, that's $\frac{7}{5} > \frac{8}{6}$.

4. Back · Table [LEC] Monthly precipitation (fictitious data).

Month	Jan.	Feb.	March	Apr.
Rain (mm)	280	300	240	120

Which month is the wettest? What is the cumulative amount of these 4 months?

Rep. February (300 mm); 940 mm.

5. Red thread · Table → formula [REG] x : $-2, -1, 0, 1$ and $f(x)$: 4, 1, 0, 1. Is the function refined? A formula.

Rep. No (variations in non-constant steps). $f(x) = x^2$ It's okay.

Test: f $(1) - f(0) = 1$ but f $(0) - f(-1) = -1$.

Form 36 March 26 Jan 2027 Equations and inequiments of 1er degree; $x^2 = a$; zero product

1. Warming · Flash [EQ] Solve 3 $x - 5 = 7$.

Rep. $x = 4$.

2. Heart · Method [EQ] Solve 2 $x + 3 \leqslant 7 - x$.

Rep. $3x \leqslant 4$: $x \leqslant \frac{4}{3}$; $S = \left]-\infty; \frac{4}{3}\right]$.

3. Heart + · Method [EQ] Solve $(x-2)(2x+6) = 0$ then $x^2 = 7$.

Rep. Product zero: $x = 2$ or $x = -3$; $x^2 = 7$: $x = \sqrt{7}$ or $x = -\sqrt{7}$.

4. Back · Flash [CMP] Compare 1.4 and $\sqrt{2}$ without calculator.

Rep. $1{,}4^2 = 1{,}96 < 2$ therefore $1{,}4 < \sqrt{2}$ (the square function is increasing on $[0;+\infty[$).

5. Red thread · Flash [GEO] ABC is isocele in A with $\widehat{BAC} = 40°$. What is the measure of $\widehat{ABC}$?

Rep. $(180 - 40) \div 2 = 70°$.